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Concept: The Heart of Unbroken Edge

Project Brief: Unbroken Edge is a disruptive innovation that redefines personal safety as a discreet, adaptive, and empowering protective system. It moves away from the traditional concept of standalone "gear" to offer a modular and invisible layer integrated directly into everyday urban clothing. The project aims to shift the entire safety ecosystem from optional, intentional-use equipment to built-in personal protection that is "always-on".

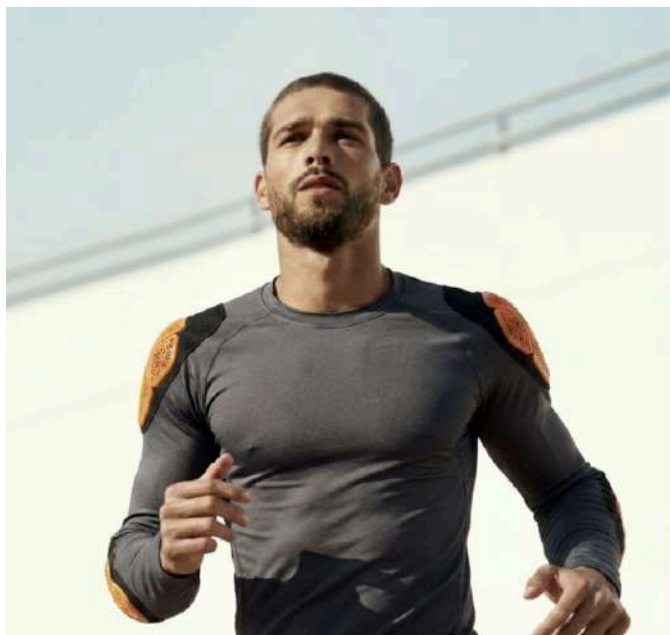
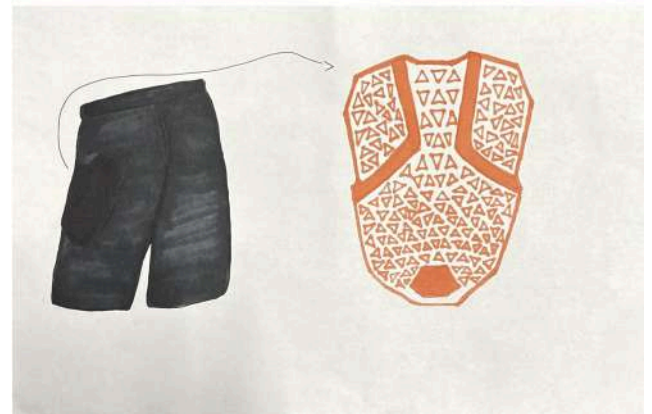
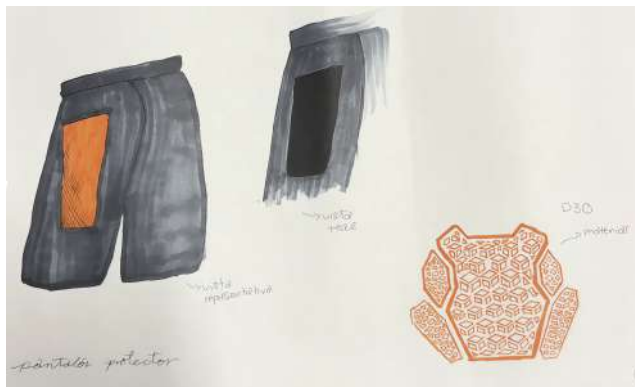
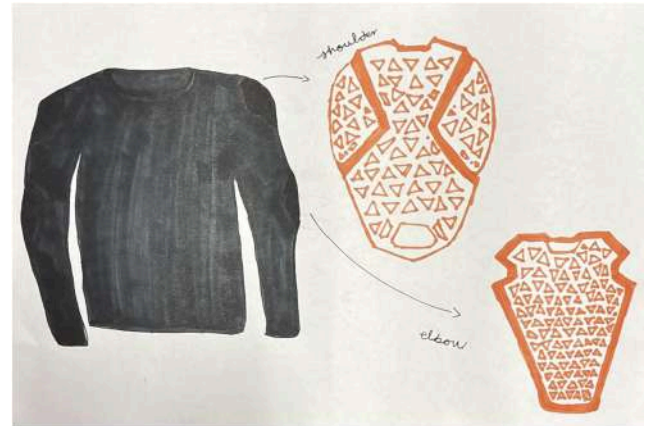
Problem Definition: The project addresses the critical lack of comfortable and aesthetic solutions for preventing bumps, scrapes, and serious injuries within the skating community.

- **The Conflict:** There is a fundamental disconnect between the high protection of traditional gear, which is often bulky, uncomfortable, and non-aesthetic, and the user's desire for comfort and style.
- **Behavioral Barriers:** Skaters prioritize convenience and "flow" in their everyday decisions, often viewing traditional pads as a sign of a "beginner" or as a physical restriction to their movement.
- **Systemic Failure:** Most existing solutions assume users will stop to "gear up" intentionally, which fails to account for the spontaneous nature of modern skating.

User Analysis: The primary target audience consists of young skaters (teenagers and young adults) who are active, trend-aware, and deeply value self-expression.

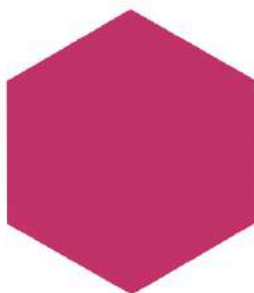
- **Profile:** These users reject anything that makes them look "bulky" or interrupts their personal identity.
- **Spontaneous Behavior:** Skaters often engage in "short bursts" of activity, skating between classes, while commuting, or just hanging out, rather than planned, dedicated training sessions.
- **Priority Shift:** For this demographic, social perception and ease of movement are more important than theoretical safety. They require a system that is frictionless and does not rely on a conscious decision to be safe.

Ideation and Sketching: The ideation phase focuses on the mantra "Not Gear. Not Armor. Clothing", translating safety into a lifestyle product.





Moodboard and Visual Language: Urban aesthetic, color palette (Vivid Pink, Oxford, Grullo, Platinum), and brand typography (N27, Helvetica).



#E71976
R231, G25, B118
C3, M98, Y23, K0
Vivid pink



#262626
R38, G38, B38
C71, M65, Y64, K69
Oxford



#A5A396
R165, G163, B134
C37, M26, Y50, K1
Grullo



#E9E4DC
R233, G228, B220
C8, M7, Y11, K0
Platinum

Brink

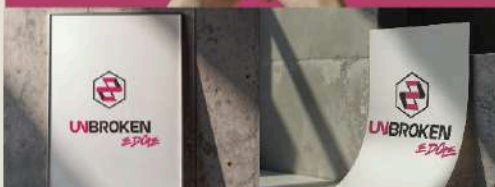
ABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789 !'#\$%&'()*=

N27

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abcdefghijklmnopqrstuvwxyz
0123456789

Heveltica Neue

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz
0123456789





Session 2 Report: Concept and Requirements

1. Project Overview

Unbroken Edge is a disruptive innovation that redefines personal safety by integrating a **modular, discreet, and adaptive** protective system directly into everyday urban clothing. Rather than being a standalone piece of "gear," it is a living system of protection that grows and adapts with the user. It aims to shift the ecosystem of safety from optional, intentional-use equipment to **built-in personal protection** that is socially accepted and "always-on".

2. User and Context

- **Primary User:** Young skaters (teenagers and young adults) who prioritize self-expression, identity, and freedom of movement.
- **User Context:** Spontaneous urban mobility. Skaters often engage in "short bursts" of activity during commutes or while hanging out, rather than planned, dedicated sessions.
- **User Behavior:** Currently, skaters reject traditional protective gear because it is perceived as bulky, restrictive, and aesthetically unpleasant, and is often associated with "beginners".

3. Problem Statement

The central problem is the **lack of a comfortable, aesthetically pleasing solution for preventing bumps and scrapes** in the skating community. Existing solutions force a choice between high protection (bulky and uncomfortable) or comfort/style (limited protection). This leads skaters to stop using safety equipment entirely to maintain their identity and freedom of movement. As the brand states: **"Falling is part of the game; getting injured... shouldn't be"**.

4. Value Proposition

Unbroken Edge offers **invisible protection** integrated into stylish, everyday wear. This allows skaters to stay safe without sacrificing their personal identity or comfort. The **modular system** provides the "free use of will" to customize protection levels based on the specific needs of an activity, making safety effortless and frictionless.

5. Concept Direction

- **"Bridging the Gap":** The project focuses on the intersection of Safety + Style, Function + Identity, and Protection + Performance.
- **"Not Gear. Not Armor. Clothing":** The fundamental shift is treating protection as a feature of the garment itself, not an external addition.
- **Passive Protection:** The system is designed to be "always-present," requiring zero decision-making from the user before a session.

6. Technical Direction (Manufacturer Report)

- **Core Material:** Utilization of **D3O technology**, a high-performance material that remains flexible until impact, allowing for a low-profile fit.
- **Modular Integration:** Garments must feature internal compartments and pockets (often using lycra layers) to secure removable pads at critical points like the spine, shoulders, ribs, hips, and knees.
- **Biomimicry:** Exploring natural structures (e.g., armadillo-inspired spine protection) for flexible yet robust defense.
- **Attachment Systems:** Focus on durability and ease of pad insertion/removal to ensure the "modular" nature remains functional over time.



7. Visual / Commercial Direction (User-Facing Report)

- **Brand Identity:** A modern, urban aesthetic using a bold palette: **Vivid Pink, Oxford, Grullo, and Platinum.**
- **Typography:** Professional and clean fonts such as **N27** and **Helvetica Neue** to reflect technology and style.
- **Commercial Positioning:** Marketed through **e-commerce** and **selected skate shops** to maintain community credibility.
- **Empowerment:** The visual language should empower the user, framing protection as a "discreet" advantage rather than a sign of vulnerability.

8. Requirements Table

Requirement	Category	Why is it important?	Technical Success Criteria	Visual / Commercial Translation	Evidence Needed
Impact Absorption	Technical	Prevent injuries from bumps and scrapes.	Effective use of D3O or equivalent smart materials.	"Protection + Performance".	Impact test data.
Invisibility / Low Profile	Design	Users reject gear that makes them look "bulky".	Pads must be thin enough to hide under standard urban fit.	"Not Gear. Not Armor. Clothing.".	Side-by-side comparison photos.
Modular Customization	Functional	Allows users to choose protection based on activity.	Secure, removable attachment systems for pads.	"Adaptable and empowering".	User testing on pad swap speed.
Freedom of Movement	Ergonomic	Vital for skating performance and "flow".	Padding placement must respect joint articulation.	"Function + Identity".	Range-of-motion study.
Breathability	Material	Skaters are active in urban environments.	To be defined.	To be defined.	Fabric breathability specs.
Aesthetic Integration	Commercial	To maintain style and social acceptance.	Adherence to brandbook colors and urban styles.	"Safety + Style".	Market survey feedback.
Durability	Technical	Ensures protection remains reliable over time.	To be defined.	Long-lasting safety.	Fatigue testing on pockets.
Passive Protection (By Default)	Functional / Behavioral	Skaters often skate spontaneously and impulsively without planning sessions; the system must protect them without requiring a conscious decision.	Protection must be integrated permanently into the base garment, eliminating the need to "put on" extra gear.	"Safety without effort" / "Always active".	Spontaneous use of behavior studies.
Anatomical Scalability	Design / Technical	The system must be capable of expanding to protect other critical areas identified in sketches, such as the spine, head, and ankles.	Compatibility of protection modules across various garments including shirts, pants, beanies, and durags.	"Full-body adaptable protection".	Prototype accessories like Durag+, Beanie+, and spine protection.
Cultural Identity Preservation	Commercial / Social	Traditional equipment is rejected because it labels the user as a "beginner"; the system must	Total alignment with current urban and street aesthetics so it does not look like safety gear.	"Bridging the gap" between safety and style.	Brand perception and social acceptance surveys.

		validate the experienced skater's identity.			
Attachment System Security	Technical / Ergonomic	Protection inserts (D3O or similar) must remain in their exact place during complex movements or multiple impacts.	Design of high-resistance internal pockets or closures that secure the module without adding visual bulk.	"Protection that stays put".	Fatigue testing on pockets and attachment systems.
Responsive Adaptability (Smart Materials)	Material / Technological	In future scenarios, materials must react not only to impact but to the user's behavior and environment.	Utilization of polymers that adjust protection levels automatically based on the intensity of the activity.	"Intelligent and living protection".	Technical specifications of real-time reactive materials.

9. MVP Definition

The **Minimum Viable Product** will consist of a **modular clothing kit** featuring one upper garment (e.g., the *Plate Jacket* or *Street Shield*) and one lower garment (e.g., *Paded Jorts*). These items will include integrated internal pockets and a standard set of removable **D3O pads** for critical impact zones. The MVP must demonstrate the **"invisible"** nature of the protection and the ease of the **modular** swap system.

10. Missing Information

- **Specific Textile Selection:** Needs evidence on the exact fabrics (denim, cargo, synthetic) for optimal durability and breathability.
- **Dimensioning:** Technical drawings with exact measurements for the "Plate" and "Pad" inserts are **to be defined**.
- **Price Target:** Detailed cost analysis and retail pricing are **to be defined**.
- **Regulatory Compliance:** Evidence needed regarding safety certifications for protective clothing.

11. Next Steps

1. **Technical Drawing:** Finalize detailed blueprints and dimensions for pad compartments in the top and bottom garments.
2. **Material Sourcing:** Identify specific suppliers for D3O and specialized textiles.
3. **Prototyping:** Create a physical MVP to conduct ergonomic and "invisibility" tests with real skaters.
4. **E-commerce Development:** Prototype the online sales platform as the primary revenue stream

Session 3 Materials

1. Material Strategy

The strategy focuses on a **hybrid apparel production process** that integrates high-performance textiles and advanced impact-protection materials into everyday streetwear. The objective is to move from visible, standalone gear to a **behavior-driven, integrated system** where protection is "invisible" and socially accepted. This involves a **modular approach**, allowing users to customize their level of safety without sacrificing their personal identity or freedom of movement.

2. Main Parts and Possible Materials

- **Base Layer:** High-stretch, breathable **spandex or elastane blends** to ensure a tight, comfortable fit that adapts to the body during movement.
- **Protective Inserts:** **D3O impact protection technology**, a non-Newtonian material that remains soft and flexible during regular use but hardens instantly upon impact.
- **Integration System:** **Sewn-in compartments or laminated pocket systems** designed to keep protective elements fixed in place.
- **Outer Garments:** Standard streetwear textiles such as **denim (for jorts/jeans)** and **heavy-duty cotton or synthetic blends (for cargo pants and jackets)** [23, Presentation visuals].



3. Material Properties Needed

- **Flexibility and Comfort:** Materials must allow for total **freedom of movement**, essential for skating "flow".
- **Breathability and Moisture-Wicking:** The base layer needs to manage perspiration during active sessions.
- **Friction Durability:** Outer fabrics must be resistant to repeated movement and urban wear-and-tear.

- **Low Profile:** Materials must be thin enough to remain **nearly invisible** underclothing to maintain fashionability.

4. Possible Production Processes

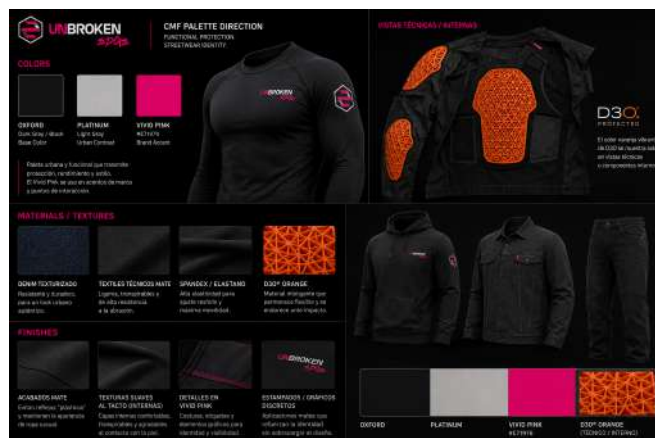
The manufacturing of the garments follows a multistep, specialized workflow:

1. **Fabric Cutting:** Using precision-cut panels for a technical fit.
2. **Base Construction:** Stitching the primary garment structure.
3. **Impact Integration:** Insertion of D3O padding into **sewn-in compartments or laminated systems**.
4. **Specialized Seams:** Utilization of **flatlock or bonded seams** to minimize bulk and improve ergonomic comfort.

5. CMF Palette Direction

The visual identity is defined by a bold, urban palette:

- **Vivid Pink (#E71976):** High-energy accent color for brand disruption.
- **Oxford (#262626):** Primary dark gray base for a solid, urban look.
- **Grullo (#A5A386) and Platinum (#E9E4DC):** Secondary technical tones for sophisticated detailing.
- **Finishes:** Textiles should feature **matte and low-key finishes** to ensure the protection remains discreet and "low-key". The D3O material itself features a characteristic **orange, geometric lattice structure**.



6. Material Risks

- **Complexity and Cost:** The integration of smart materials results in **complex design and higher production costs** compared to basic gear.
- **Skepticism of Performance:** Users may doubt the effectiveness of "invisible" safety compared to bulky traditional pads.

- **Durability and Maintenance:** The long-term performance of integrated pads after repeated **wash cycles** needs further evidence.

7. Alternative Materials

While D3O is the primary focus, the sources suggest potential alternatives or past inspirations:

- **Integrated Soft Paddings:** Similar to those used in traditional football pants.
- **Polyethylene Pads:** Mentioned as a common existing material in undergear.
- **Compression Wear Fabrics:** Suggested as a baseline for the base layer.



ALTERNATIVE MATERIALS

INNOVATIVE SOLUTIONS FOR IMPACT PROTECTION

1. BIOMIMICRY STRUCTURES

Estructuras inspiradas en escamas de armadillo para protección mecánica flexible.



CARACTERÍSTICAS

- Flexibilidad multidireccional.
- Alta resistencia al impacto.
- Adaptabilidad al movimiento.
- Ventilación entre segmentos.

USO RECOMENDADO
Hombros, codos, espalda.



2. ENCAPSULATED GELS

Gel de absorción de impacto para zonas de menor riesgo.



CARACTERÍSTICAS

- Excelente absorción de impacto.
- Suave y cómodo.
- Ligero y delgado.
- Se adapta al cuerpo.

USO RECOMENDADO
Caderas, muslos, costillas.



3. INFLATABLE AIR-BANDS

Anillos de aire o cojines inflables para protección craneal.



CARACTERÍSTICAS

- Absorción de impactos por aire.
- Ajustable e inflable.
- Muy ligero y compacto.
- Fácil de almacenar y transportar.

USO RECOMENDADO
Cabeza (uso bajo gorras o integrado en accesorios).



4. SOFT FOAMS

Espumas de poliuretano de baja densidad como alternativa económica en zonas no críticas.



CARACTERÍSTICAS

- Ligera y económica.
- Cómoda y flexible.
- Absorción básica de impacto.
- Fácil de cortar y adaptar.

USO RECOMENDADO
Áreas no críticas, refuerzos secundarios y comodidad.



COMPARATIVA DE MATERIALES	ABSORCIÓN DE IMPACTO	FLEXIBILIDAD	COMODIDAD	PESO	VENTILACIÓN	COSTO
1. BIOMIMICRY STRUCTURES	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
2. ENCAPSULATED GELS	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
3. INFLATABLE AIR-BANDS	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
4. SOFT FOAMS	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★

8. How to Explain Materials in the Technical Report

The technical report should emphasize **performance engineering and validation**. It must detail the **impact resistance testing** and the science of **non-Newtonian material behavior** (D3O). Descriptions should focus on "functional value," "hybrid production," and "precision-cut panels" to demonstrate how the materials "bridge the gap" between comfort and maximum protection.

9. How to Communicate Materials in the Visual/Commercial Report

The commercial narrative must prioritize **style and empowerment**. Communicate materials through slogans like **"NOT GEAR. NOT ARMOR. CLOTHING"**. Highlight the **"invisible"**



and **"always-on"** nature of the safety system to appeal to the user's desire for effortless style. Use visual storytelling to show how the materials **"disappear"** into everyday life while providing the confidence to skate without fear



SHOPTECHNOLOGYABOUTJOURNALCONTACT



SAFETY + STYLE

**GETTING INJURED...
SHOULDN'T BE.**

**NOT GEAR. NOT ARMOR.
CLOTHING.**

SHOP NOW →



01 SAFETY + STYLE

02 FUNCTION + IDENTITY

03 PROTECTION + PERFORMANCE

**ADVANCED PROTECTION**
High impact areas.
Discreetly protected.

**LIGHTWEIGHT DESIGN**
Move freely.
Zero bulk.

**EVERYDAY STYLE**
Looks like streetwear.
Performs like protection.

**BUILT FOR SKATERS**
Tested by skaters.
Trusted on impact.

SHOP

**PROTECTION
YOU WEAR.**

Technical skatewear with
impact protection built in.
For the ones who push
the limits.

SHOP ALL →

**HOODIE**
● ● \$169.00

**PANTS**
● ● \$156.00

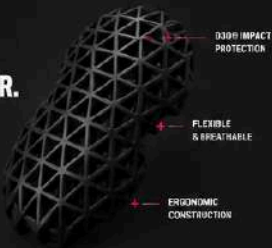
**LONG SLEEVE**
● ● \$99.00

TECHNOLOGY

**BUILT DIFFERENT.
PROTECTS BETTER.**

D3O® impact protection.
Ergonomic fit. Maximum
flexibility. Zero compromise.

LEARN MORE →

**D3O® IMPACT
PROTECTION**

**FLEXIBLE
& BREATHABLE**

**ERGONOMIC
CONSTRUCTION**

JOURNAL

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THE STREETS.**

Skate culture. Real talk.
Injury prevention. Progression.
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**HOW TO SKATE HARD
AND STAY IN THE GAME**
MAY 12, 2024



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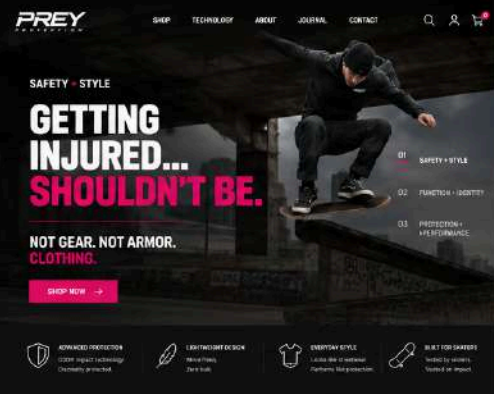
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SAFETY + STYLE

GETTING INJURED... SHOULDN'T BE.

NOT GEAR. NOT ARMOR. CLOTHING.

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BUILD YOUR PROTECTION

Our modular system lets you choose your protection. You're in control.

 SHOULDERS D30® Ghost Level 1

 ELBOWS D30® Ghost Level 1

 SPINE D30® Viper Level 2

 HIPS D30® Ghost Level 1

 ELBOW PROTECTION D30® Ghost Level 1

Lightweight. Flexible. Moves with you.

REMOVE

PROTECTION LEVEL

- Level 1 / Light
- Level 2 / Medium
- Level 3 / Heavy

[RESET BUILD](#)

SHOP

TOPS BOTTOMS HEADWEAR SOCKS ACCESSORIES



STREET SHIELD HOODIE

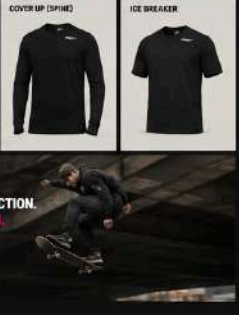
\$169.00

Technical hoodie with modular D30® protection. Streetwear look. Pre-knit protection.

COLOR: BLACK

SIZE: S M L XL XXL

[ADD TO CART](#)



MODULAR PROTECTION. INVISIBLE DESIGN.

[SHOP TIPS](#)

D30® INTELLIGENT IMPACT PROTECTION



SOFT & FLEXIBLE
Moves with you. Comfortable about any activity.

LOCKS ON IMPACT
Microcaps lock together on impact to absorb and dissipate energy.

RETURNS TO FLEXIBLE
After impact, D30® returns to its soft, flexible state.

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TOPS **BOTTOMS** HEADWEAR SOCKS ACCESSORIES



PADDED JORTS

\$129.00



CARGO

\$159.00



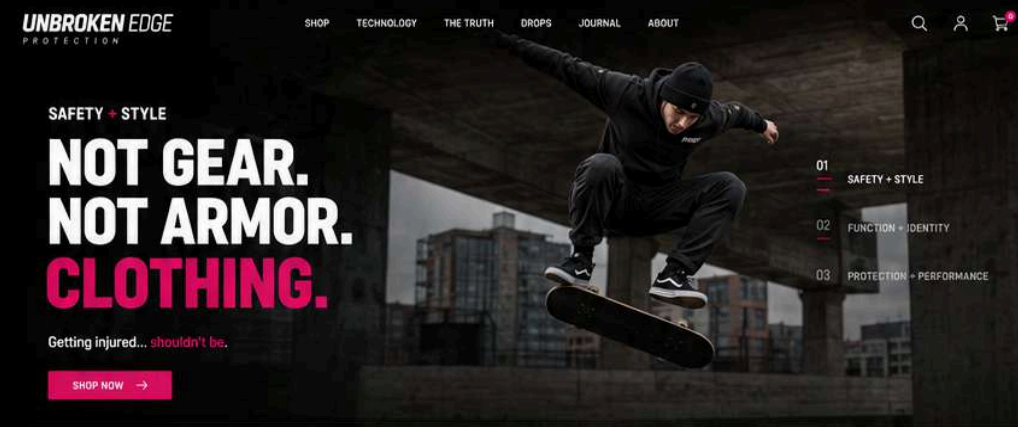
COMPRESSION SOCKS

\$25.00

MODULAR SYSTEM
Add on modular protection where you need it.

URBAN FIT
Streetwear look. Protection inside.

BUILT TO MOVE
Designed for skating. Made for anywhere.



SAFETY + STYLE

NOT GEAR. NOT ARMOR. CLOTHING.

Getting injured... shouldn't be.

[SHOP NOW](#)

BUILD YOUR PROTECTION

Our modular system lets you choose your protection. You're in control.

 SHOULDERS D30® Ghost Level 1

 ELBOWS D30® Ghost Level 1

 SPINE D30® Viper Level 2

 HIPS D30® Ghost Level 1

 ELBOW PROTECTION D30® Ghost Level 1

Lightweight. Flexible. Moves with you. Discreet.

REMOVE

PROTECTION LEVEL

- Level 1 / Light
- Level 2 / Medium
- Level 3 / Heavy

[RESET BUILD](#)

THE TRUTH**US VS. THEM****THE PROBLEM**

Bulky. Restrictive.
Beginner-looking.
Traditional gear wasn't
made for skaters.

**THE SOLUTION**

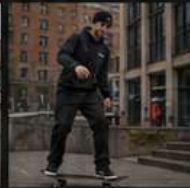
Unseen protection.
Unlimited freedom.
Designed for skating.
Built for culture.

[SEE THE DIFFERENCE →](#)**SKATERS. CITIES. CULTURE.****BUILT WITH THE CULTURE**[VIEW LOOKBOOK →](#)

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LIMITED DROPS. MAXIMUM IMPACT.**UPCOMING DROP****URBAN SHIELD
COLLECTION**

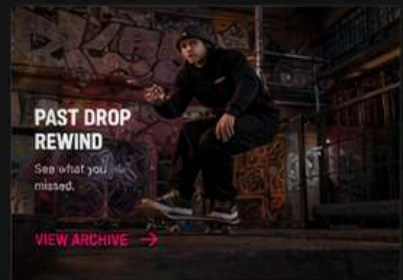
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DAYS HOURS MINS SECS

**PAST DROP
REWIND**

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missed.

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IMPACT PROTECTION****MODULAR SYSTEM
CHOOSE YOUR PROTECTION****DISCREET DESIGN
LOOKS LIKE STREETWEAR****BUILT FOR SKATERS
TESTED. TRUSTED.****FREE SHIPPING
ON ORDERS \$100+****UNBROKEN EDGE
PROTECTION**

Not gear. Not armor.
Clothing with intelligent protection.
For skaters. By skaters.

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Testing
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Session 4 Report: Technical Drawings, Exploded View, and Preliminary BOM

1. Product Architecture

The **Unbroken Edge** architecture is a **modular streetwear-integrated system**. It is composed of a hybrid assembly where a technical **base layer** (hermetic spandex/elastane) is integrated into or worn beneath standard **urban garments** (outer shell). The architecture is designed to accommodate **D3O impact modules** within discreet internal compartments, allowing for a customizable level of protection that remains invisible to the public.

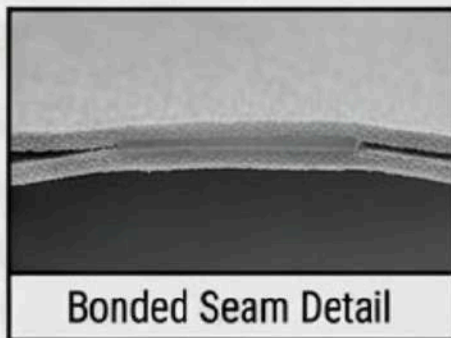
2. Parts Breakdown

- **Outer Shell:** Streetwear-style garments like pants (*Padded biker shorts*), and shirts (*Plate shirt*).
- **Base Layer System:** A hermetic spandex/elastane structure connected or sewn into the clothes.
- **Protection Modules:** Removable D3O impact inserts for elbows, shoulders, lateral thighs, and spine.
- **Attachment Mechanisms:** Laminated pocket systems or sewn-in internal compartments.
- **Connection Features:** Detachable sleeves for top garments to allow for user customization.

3. Manufactured Parts

- **Integrated Garment Shells:** Outer clothing designed with specific internal housing for protection.
- **Internal Spandex Sleeves:** Custom-cut elastic base layers for pad retention.
- **Laminated Retention Pockets:** Specialized internal compartments for module security.
- **Custom Seams:** Flatlock or bonded seam finishes for low-profile comfort

CUSTOM SEAMS: LOW-PROFILE COMFORT



4. Commercial Parts

- **D3O Impact Inserts:** Advanced non-Newtonian polymer modules supplied by D3O.
- **Performance Textiles:** High-stretch spandex, breathable elastane blends, and reinforced denim.
- **Fasteners/Closures:** Components for the detachable sleeve system [needs evidence].

5. Exploded View Description

The exploded view illustrates the **three-layer vertical integration** of the product:

1. **Level 1 (Top):** The **Outer Shell** (e.g., denim jacket or cargo pants) showing the urban aesthetic.
2. **Level 2 (Middle):** The **Impact Modules** (D3O inserts) extracted from their specific anatomical locations (elbows, shoulders, thighs).
3. **Level 3 (Bottom):** The **Base Layer and Retention System**, showcasing the internal hermetic spandex and the laminated pocket structures that hold the modules in place. *The view also highlights the detachable sleeves separated from the main torso of the upper garments.*

EXPLODED-VIEW: ADVANCED URBAN PROTECTIVE GEAR





6. Drawing Checklist

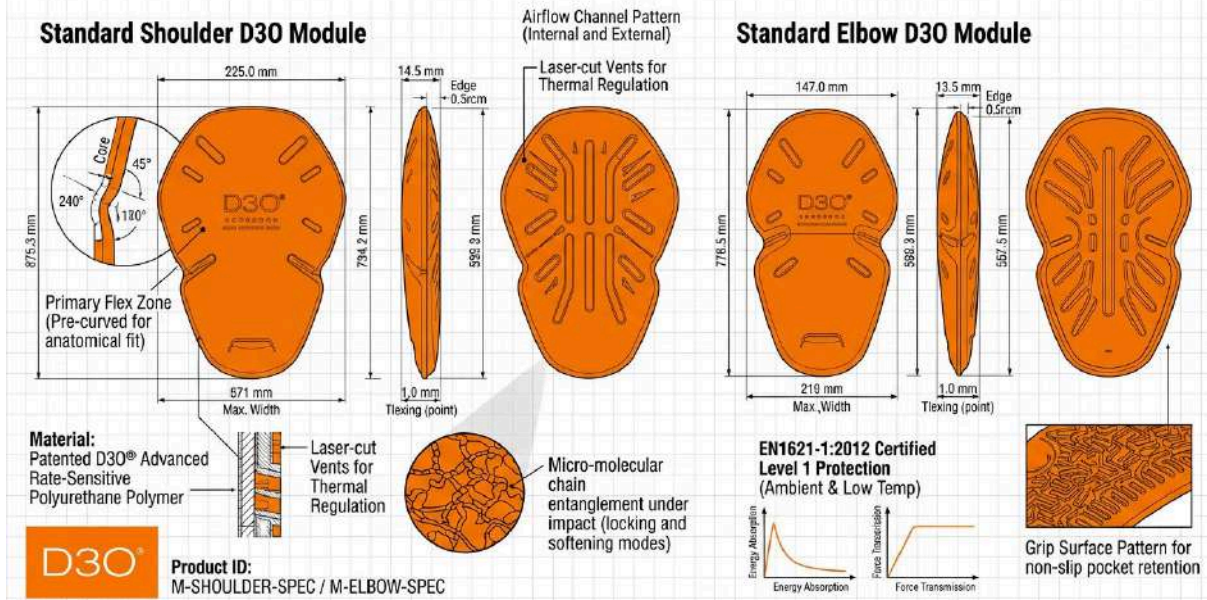
- [] **Full Assembly Drawing:** Showing the integrated garment with modules hidden.
- [] **Exploded View Drawing:** Detailing the layer separation.

EXPLODED-VIEW: ADVANCED URBAN PROTECTIVE GEAR



- [] **D30 Module Detailed Drawings:** Specifications for standard elbow/shoulder inserts.

D30 MODULE DETAILED SPECIFICATIONS: STANDARD INSERTS



- [] **Pocket Construction Detail:** Section view of laminated or sewn-in compartments.
- [] **Detachable Sleeve Mechanism:** Detail of the connection interface.

EXPLODED-VIEW: ADVANCED URBAN PROTECTIVE GEAR



7. Preliminary BOM (Bill of Materials)

Part / Component	Quantity	Type	Material	Process or Supplier	Estimated Cost	Function	Status
Outer Garment Shell	1 per unit	Manufactured	Reinforced Denim/Cotton	To be defined	To be defined	Urban style & outer protection	Draft
Base Layer (Spandex)	1 per unit	Manufactured	Spandex / Elastane blend	Precision Fabric Cutting	To be defined	Fit & pad support	Draft
D3O Impact Inserts	Set (2-4)	Commercial	D3O Polymer	D3O (Supplier)	To be defined	Impact absorption	Finalized
Internal Pockets	Set (2-4)	Manufactured	High-tenacity mesh/film	Lamination / Sewing	To be defined	Module retention	Needs testing
Detachable Sleeves	1 pair	Manufactured	Spandex/Denim blend	To be defined	To be defined	Modular customization	Draft
Seam Finishing	N/A	Process	N/A	Flatlock / Bonding	To be defined	Comfort & low profile	Finalized

8. Assembly Sequence

- Fabric Cutting:** Precision cutting of panels for both the outer shell and the internal spandex base.
- Base Assembly:** Stitching the primary garment structure together.
- Pocket Integration:** Lamination or sewing of the internal protective zones and pocket systems.

4. **Padding Insertion:** Integration of the D3O modules into the designated compartments.
5. **Final Finishing:** Application of flatlock or bonded seams for a seamless, "invisible" finish.
6. **Quality Control:** Conduct impact, wash-cycle, and user wear testing.

9. Technical Risks

- **Attachment Fatigue:** The retention pockets may stretch or fail after repeated use and washing, causing modules to shift.
- **Thermal Regulation:** The hybrid layering (spandex + outer garment + polymer pads) may lead to excessive heat retention during active use [needs testing].
- **Production Complexity:** The multistep process involving lamination and precision spandex integration is more complex than standard apparel.
- **Material Cost:** High reliance on D3O technology may affect final price points and scalability.

10. Missing Information

- **Specific Dimensions:** Precise measurements for pad sizes across different garment sizes [needs measurement].
- **Sleeve Mechanism:** Technical details of the detachable sleeve fastening system (zippers, snaps, or velcro) [needs evidence].
- **Durability Data:** Performance of laminated pockets after 50+ wash cycles [needs testing].
- **Supplier Costs:** Final quotes for textile manufacturing and wholesale D3O components [to be defined].

11. Next Steps

- **Functional Prototyping:** Create physical samples to test real-world fit and module stability.
- **User Testing:** Gather feedback from skaters on "flow" and freedom of movement.
- **Manufacturing Partnerships:** Establish agreements with clothing manufacturers capable of handling hybrid textile production.
- **Certification:** Pursue third-party impact performance certifications to build user trust.



Session 5 Report: Ergonomics, Use Diagrams, and Validation Plan

1. User Interaction

The user interacts with **Unbroken Edge** as a high-performance **2-piece spandex set**, consisting of a **long-sleeve shirt and shorts**. Unlike traditional gear that requires straps or external adjustments, this product is worn as a "second skin" base layer. The user simply pulls on the garments, which stay in place due to the high-stretch elastane blends. The protection pads are **secured permanently via thermo-adherent technology** at the shoulders, elbows, and hips, meaning the user does not need to manually insert or adjust the protection before use.

2. Use Sequence

1. **Dressing:** The user puts on the long-sleeve spandex shirt and the spandex shorts.
2. **Integration:** Because the system is **low-key and nearly invisible to the eye**, the user can wear their preferred streetwear over the set without any visible bulk.
3. **Spontaneous Activity:** The user engages in skating or urban mobility. The **passive protection** is "always-on," requiring zero decision-making.
4. **Impact Event:** During a fall, the integrated **D3O pads** at the shoulders, elbows, or hips harden instantly to absorb the shock.
5. **Post-Activity:** The user continues their daily routine without needing to remove "gear," as the spandex set remains comfortable for all-day wear.

3. Ergonomic Points

- **Body Contact and Grip:** The spandex set provides continuous compression and contact, ensuring the pads remain precisely over the **critical impact zones** (shoulders, elbows, and hips) even during high-velocity movement.



- **Movement and Comfort:** The thinness of the thermo-adhered pads and the flexibility of the spandex allow for a total **freedom of movement**, essential for maintaining a skater's "flow".
- **Weight:** The lightweight nature of the 2-piece set prevents the "heavy" feeling associated with traditional reinforced jackets or pants.

4. Critical Contact Points

- **Body:** Specifically, the **shoulders and elbows** (upper garment) and the **hips** (lower garment).



- **Street Environment:** These specific points are the first to hit the ground or urban obstacles during a fall.
- **Outerwear:** The interaction between the spandex set and the user's standard clothing must be frictionless to ensure the protection remains **invisible**.

5. Safety Risks

- **Thermo-adherent Failure:** The pads could peel or shift if the thermo-adherent bond weakens over time (needs testing).
- **Movement Blocking:** If the pads are not perfectly anatomically aligned during the thermo-adherent process, they could restrict joint movement.
- **Chafing:** Internal seams or the edges of the thermo-adhered pads could cause skin irritation during long sessions.
- **Perceived Protection Gap:** Because the pads are **invisible to the view**, a user might feel less safe than when wearing bulky, visible gear until they experience a successful impact absorption.



6. Validation Plan

1. **Thermo-Adhesion Durability Test:** Subject the garments to high-stress stretches to ensure the pads remain fixed.
2. **Wash-Cycle Performance:** Specifically test the **thermo-adherent bond** and spandex elasticity after 20+ washes.
3. **Invisibility Audit:** Visual testing with streetwear worn over the set to confirm the "low-key" aesthetic is maintained.
4. **Impact Testing:** Lab verification that the fixed D3O pads at the shoulders, elbows, and hips meet safety standards.
5. **Skater Field Trials:** Real-use testing to confirm "freedom of movement" and "comfort" during impulsive skating.

7. Test Criteria

- **Adhesion:** Pads must not peel at the edges under 20% fabric elongation.
- **Invisibility:** A panel of observers should not be able to identify the pads' location when a standard T-shirt and jeans are worn over the set.
- **Ergonomics:** User feedback must consistently rate the set as "unnoticeable" or "frictionless".

8. Evidence Needed

- **Macro photography** of the thermo-adherent edges.
- **Side-by-side video** of a skater performing tricks with and without the spandex set to prove no loss of "flow".
- **Wash-test data table** tracking pad adhesion.
- **Comparison photos** demonstrating the "invisible" profile.

9. Technical Explanation

For the manufacturer, the focus must be on the **thermo-adherent application process** and the precision placement of pads on the spandex base. The technical report must specify the **exact temperature and pressure** required to secure the D3O pads to the elastane blend without damaging the fabric's moisture-wicking properties.

10. Visual / Commercial Explanation

Commercial communication should highlight that this is **"Not Gear. Not Armor. Clothing"**. It should emphasize that the user is buying a **2-piece performance set** that provides **"Superpowers"** (invisible safety) without changing their look. The message is: *"Wear it like normal clothes"*.



12. Next Steps

1. **Produce a prototype** of the long-sleeve shirt and shorts with thermo-adhered pads.
2. **Conduct initial "blind" invisibility tests** with different types of streetwear.
3. **Perform a 10-wash cycle test** to observe thermo-adherent integrity.
4. **Finalize the pattern design** for the shoulders, elbows, and hips zones.
5. **Gather feedback** on the "2-piece set" concept from the primary target group (skaters)